

Smart Cane Environmental Awareness System

IoT Term Project — First Draft Presentation

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Raspberry Pi 4 · Arducam ToF · OpenCV · Python

A. Main Idea & Objective

Problem: Visually impaired users struggle to navigate indoor environments safely and independently.

Our solution: A smart cane device that provides real-time environmental awareness through two mechanisms:

● Obstacle Detection

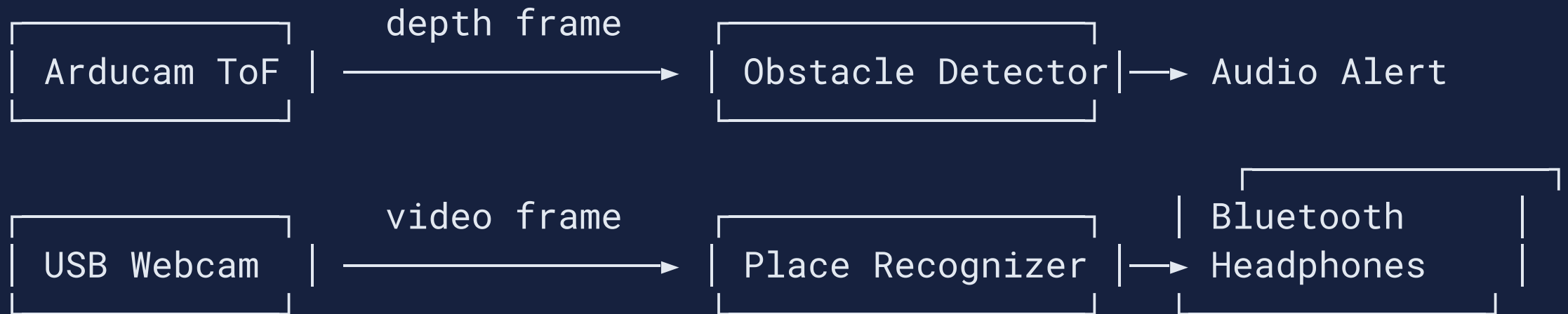
The ToF depth camera continuously monitors the forward path and warns the user about nearby obstacles through

● Location Recognition

The webcam recognizes previously trained rooms or locations and tells the user where they currently are.

B. System Design & Methodology

Core loop running at 2 Hz — no cloud, fully offline:



Concern	Approach	Frequency
Obstacle detection	Depth threshold on ToF frame	Every cycle

D. Hardware Components

Processing & Sensing

Component	Role
Raspberry Pi 4 (4GB)	Central compute
Arducam ToF B0410	Depth camera (CSI)
USB Webcam C270	Room recognition
Bluetooth Headphones	Audio output

Why this hardware?

- Runs **fully offline** — no GPU

C. Key Code — Obstacle Detection

`src/nav_assistant/obstacle/detector.py`

```
def check(self, tof_frame: ToFFrame) -> Optional[ObstacleAlert]:  
    # Get minimum depth in the central forward zone  
    min_depth = tof_frame.forward_min_depth(zone_fraction=0.33)  
  
    if min_depth > 0 and min_depth < self._threshold:  
        now = time.monotonic()  
        if now - self._last_alert_time >= self._cooldown:  
            self._last_alert_time = now  
            return ObstacleAlert(min_depth=min_depth, ...)  
  
    return None # no alert
```

C. Key Code — Location Recognition

`src/nav_assistant/localization/place_recognizer.py`

```
def recognize(self, gray_frame: np.ndarray) -> RecognitionResult:
    _, query_descs = self._orb.detectAndCompute(gray_frame, None)

    for waypoint, stored_descs in self._index:
        matches = self._matcher.knnMatch(query_descs, stored_descs, k=2)
        # Lowe's ratio test - keep only strong matches
        good = [m for m, n in matches
                 if m.distance < 0.75 * n.distance]

        if len(good) > best_good:
            best_good = len(good)
            best_wp = waypoint
```

Current Progress

Phase	Status	Description
Phase 1 — Sensors	✓ Complete	Webcam + ToF working on RPi hardware
Phase 2 — Training	✓ Complete	2 locations trained (5000 descriptors each)
Phase 3 — Awareness	🔄 In Progress	Awareness loop built, testing underway
Phase 4 — Tuning	⌚ Pending	Threshold and confidence tuning

What runs today:

E. Challenges & Solutions

Challenge	Root Cause	Solution
Wrong camera device index	USB webcam index not stable across reboots	Auto-detect by V4L2 device name at startup
ToF SDK <code>DeviceType.TOF</code> missing	API changed between SDK versions	Use <code>FrameType.DEPTH</code> , parse official examples
pyttsx3 voice error	Broken espeak-ng driver on Python 3.13	Call <code>espeak-ng</code> directly via subprocess
Webcam returns blank frames	UVC sensor warmup time	Discard first 5 frames after <code>open()</code>
Single frame per	ORB needs visual	2-minute continuous capture

F. Expected Output & Benefits

What the system says to the user:

"Awareness system started."

"You are in the office."

"Obstacle detected 0.8 metres ahead."

"You are still in the office."

"You are in the hallway."

G. Next Steps & Project Plan

Immediate (this week)

- End-to-end test of `awareness.py` on hardware
- Tune `confidence_threshold` and `min_good_matches` for reliable recognition
- Tune `alert_threshold_m` for comfortable obstacle warning distance

Short term (next 2 weeks)

Thank You

Smart Cane Environmental Awareness System



github.com/w1gg0r/smart-nav-cane

Built with Raspberry Pi 4 · Arducam ToF · OpenCV · Python

Questions?